pfi

# Python for Fixed Income

COURSE 7 · NLP, RAG, AND LLMS

# Zero-Shot Classification against TF-IDF

WEEK 2 · DAY 2

Created by **Jeff Lenamon**

This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

# Three models, one set of rows

### TF-IDF (day 3)

Fitted on 3,650 labelled sections.

### Zero-shot

No labelled section: eight descriptions the course wrote.

### Embeddings

MiniLM's 384 numbers, then a logistic regression.

**Fitted on 3,650 sections from 2019 to 2023; scored on 741 from 2024; 735 from 2025 closed.**

# Real documents, described

**An 8-K section's label is the filer's item; a model's output is the item it assigns**

| Allowed names | Never called |
| --- | --- |
| the item the model assigns | a default signal |
| the model's probability or score in this file | a distress flag or a warning |
| the filer's item | a credit event or credit view |

**Item 5.02 sections name people: counted by item only, never printed.**

# How a zero-shot classifier works

| | |
|---|---|
| **The problem** | Label text with names the model never trained on |
| **Hypotheses** | One per label, from a template, paired with the text: eight passes per section |
| **Entailment** | The model scores how well each hypothesis follows from the text |
| **Softmax** | The eight scores become shares that add up to 1; the largest is the label |
| **What you control** | The label wording, the template, multi_label; out come labels and scores |

# Natural language inference, used as a classifier

### Premise

The 8-K section, its first 120 words.

### Hypothesis, one per label

This section of a company's current report is about an event that accelerates a debt obligation.

**12,750,594**

parameters

**8**

passes per section

**6**

files, 26,457,473 bytes

**The pipeline returns the labels sorted by score, largest first.**

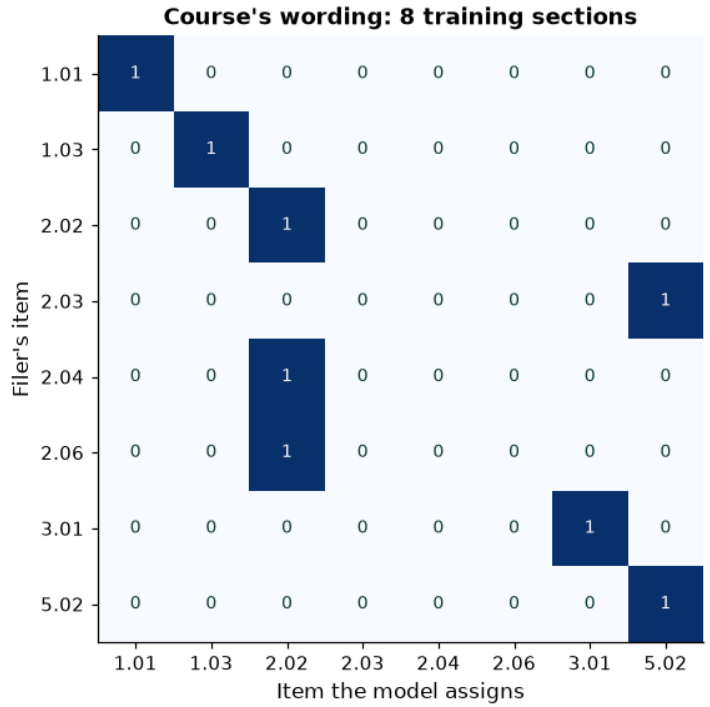
# Where the transformer comes from

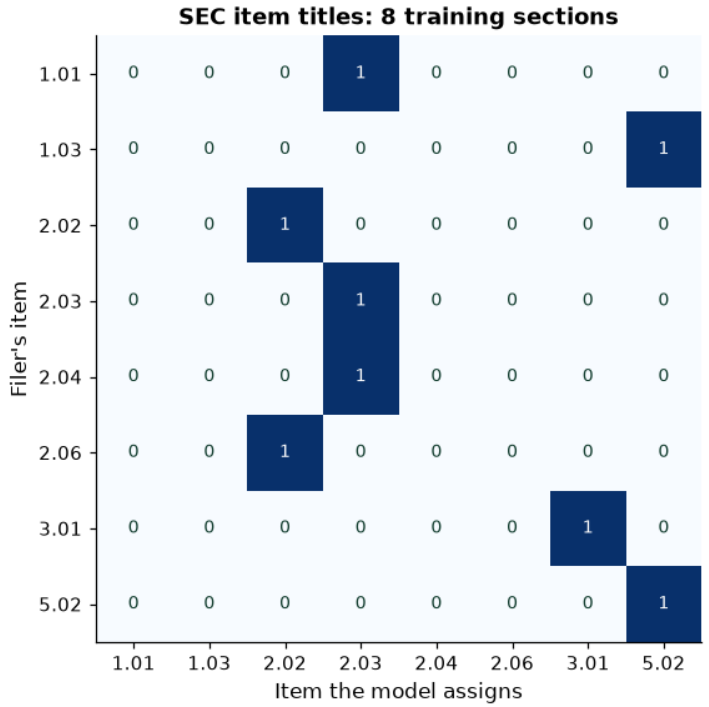
| | |
|---|---|
| **The paper** | "Attention Is All You Need": arXiv 1706.03762, submitted 2017-06-12 |
| **Authors** | Vaswani, Shazeer, Parmar, Uszkoreit, Jones, Gomez, Kaiser, Polosukhin |
| **Its abstract** | "the Transformer, based solely on attention mechanisms" |

**The course uses the model as published. The paper itself was not read.**

Source, read 8 October 2026: arxiv.org/abs/1706.03762 (its arXiv API record)

# The label wording matters




**The course's wording: 5 of 8 on the diagonal**

**The SEC's item titles: 4 of 8**

One training section of each item, drawn with the seed, 128 passes of the model. Eight is too few to choose on; it shows the words move the result.

**Label wording is fixed on training rows, never on the rows that score the model.**

# A cached output, checked live

**3,675**

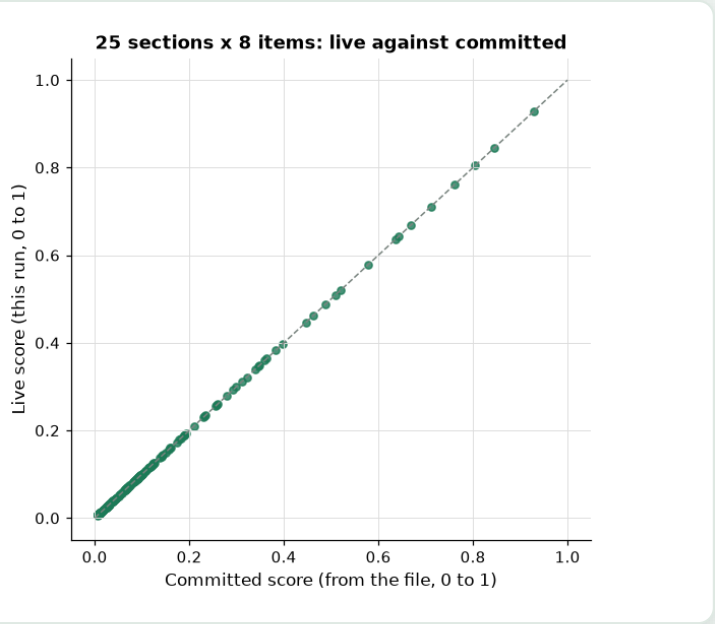
seconds: the tool on one thread

**25 of 25**

the same item live and from the file

**True**

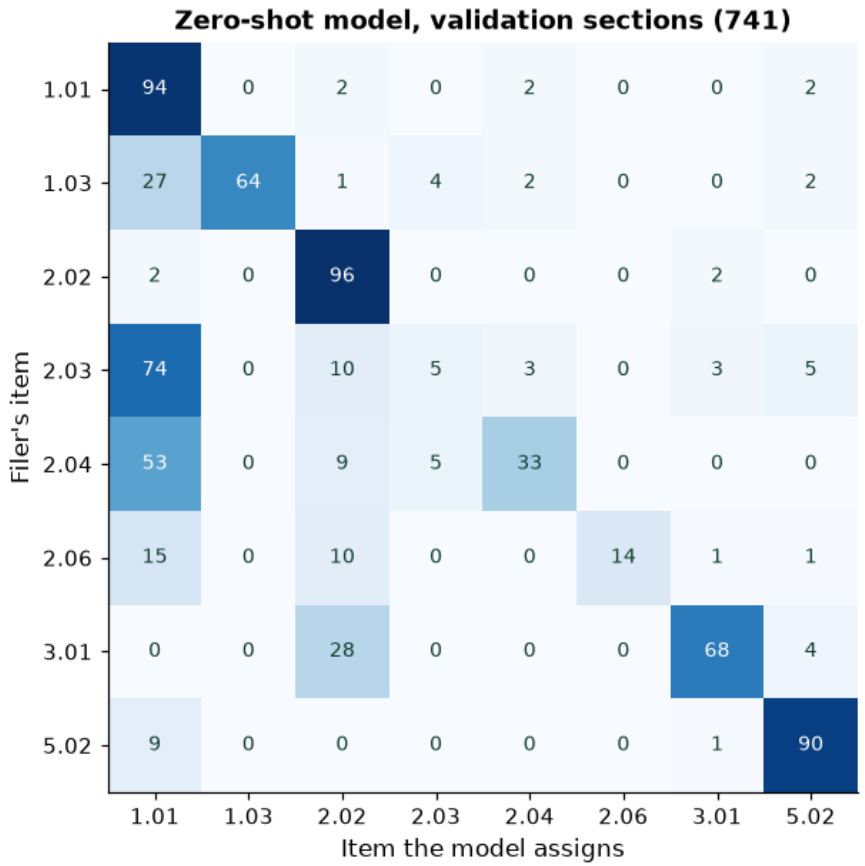
live scores equal the file's

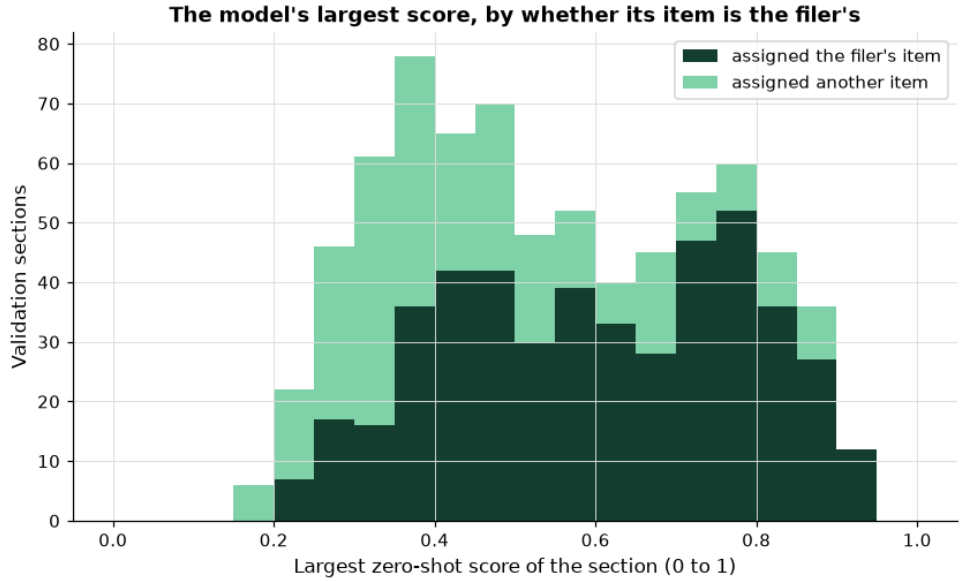



```
tolerance = 1e-4 if torch.get_num_threads() == 1 else 1e-3
# one thread: within 1e-6; default threads: up to 3e-4
```

**A live zero-shot score is never printed: scores print from the committed file.**

# Where the zero-shot model misses



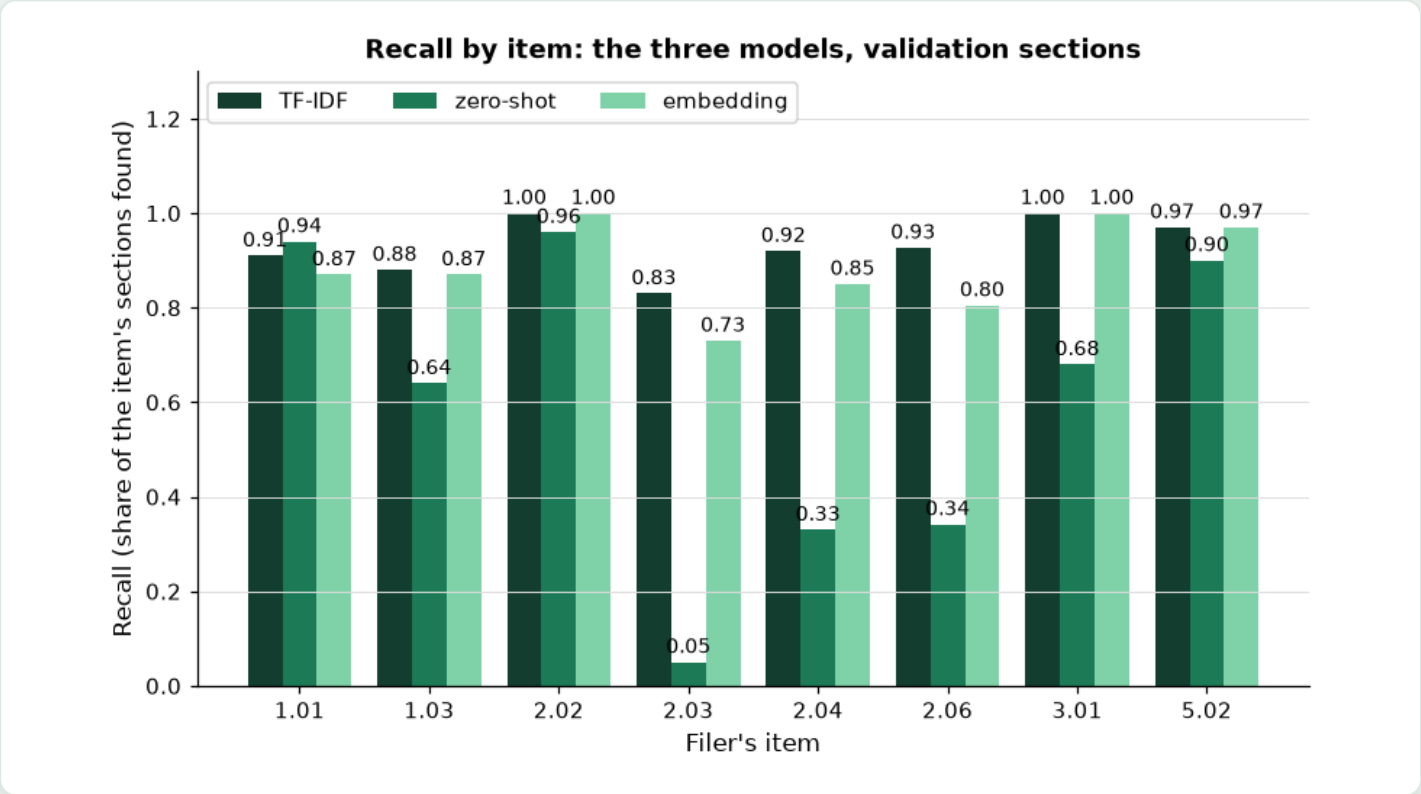




**0.3698 of the sections are assigned 1.01. A score is not a probability.**

# How embeddings become features

| | |
|---|---|
| **The problem** | Fit an ordinary classifier on text, with no word counts |
| **The table** | Each section becomes 384 numbers from day 6's model: 384 columns |
| **The fit** | A logistic regression on the training rows' vectors and their items |
| **What you control** | The model that made the vectors, the regression's settings, the rows |
| **What comes out** | An item for each section, scored like any other model |

# Three models on the validation rows



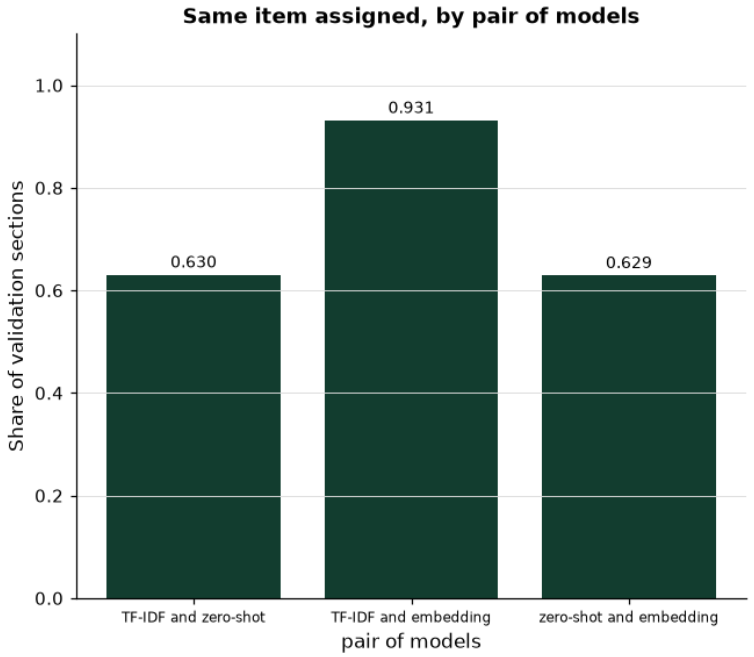

| | TF-IDF | Zero-shot | Embedding |
|---|---|---|---|
| recall 1.01 | 0.9100 | 0.9400 | 0.8700 |
| recall 1.03 | 0.8800 | 0.6400 | 0.8700 |
| recall 2.03 | 0.8300 | 0.0500 | 0.7300 |
| recall 2.04 | 0.9200 | 0.3300 | 0.8500 |
| macro F1 | 0.9271 | 0.5951 | 0.8882 |

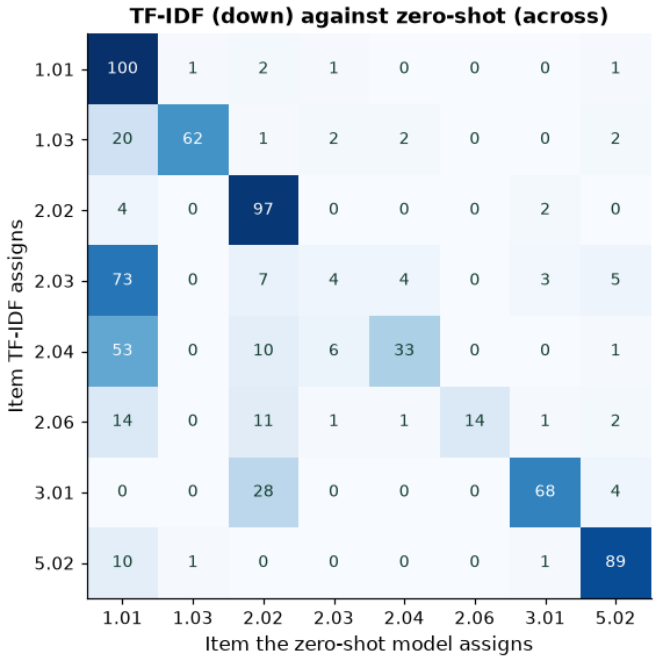
50 live embeddings equal the committed rows within 1e-6: True. The logistic regression is fitted on float64 copies.

# How agreement is counted

| | |
|---|---|
| **The problem** | Compare two models when the truth is not the question |
| **The cross** | Items one model assigns down the side, the other model's across the top |
| **Counts** | Each cell counts the sections that got that pair of items |
| **Agreement** | The diagonal's share of all the sections |
| **What it says** | How alike two models are, never which one is right |

# Where the three models agree




All three assign the same item to 452 of 741 sections; on those it is the filer's item 0.9823 of the time.

**Agreement is not accuracy: two models can agree and both be wrong.**

# A sentiment pipeline, named and not run

```
SUPPORTED_TASKS["text-classification"]["default"]   # read, nothing downloaded
# distilbert/distilbert-base-uncased-finetuned-sst-2-english, revision 714eb0f
```

Its model card: "fine-tuned on SST-2". The SST-2 card: "11,855 single sentences extracted from movie reviews".

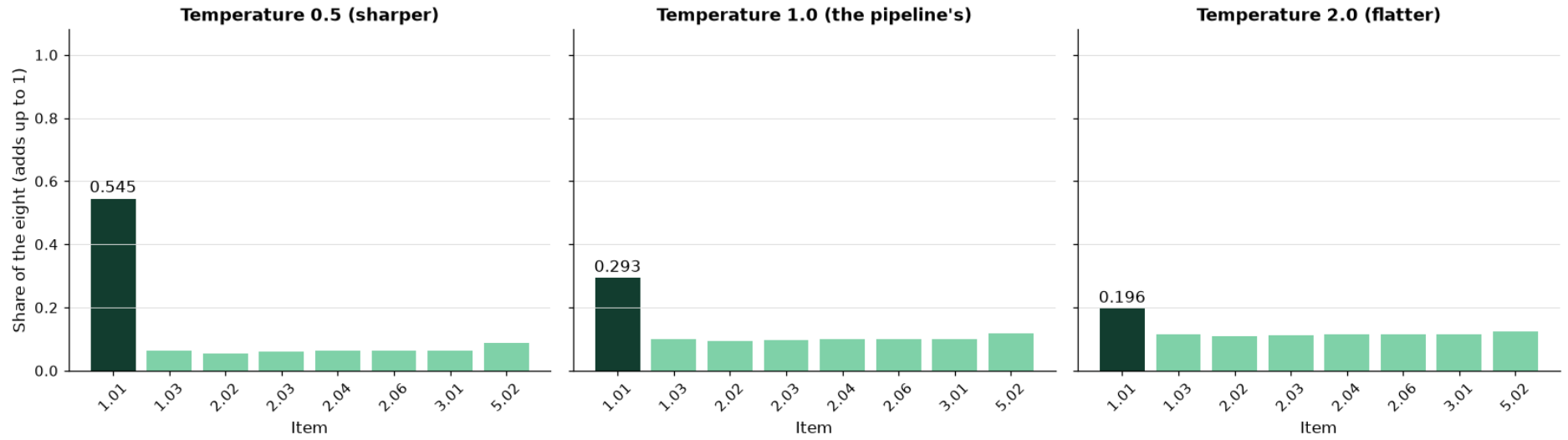
**A label learned from film reviews means nothing for an 8-K section.**

Sources, read 8 October 2026: the transformers 5.19.0 source; huggingface.co model and dataset cards

# How the zero-shot request works

| | |
|---|---|
| **The problem** | Four settings, three of which change what the scores mean |
| **Hypotheses** | The labels go through the template: the wording of both reaches every score |
| **Softmax** | multi_label=False: eight shares that add up to 1; True: each label alone |
| **Temperature** | Divide the inputs of a softmax by T: below 1 sharper, above 1 flatter |
| **What you control** | Labels, template, multi_label; the pipeline always uses T = 1 |

# What a temperature does




**0.545**

T = 0.5

**0.293**

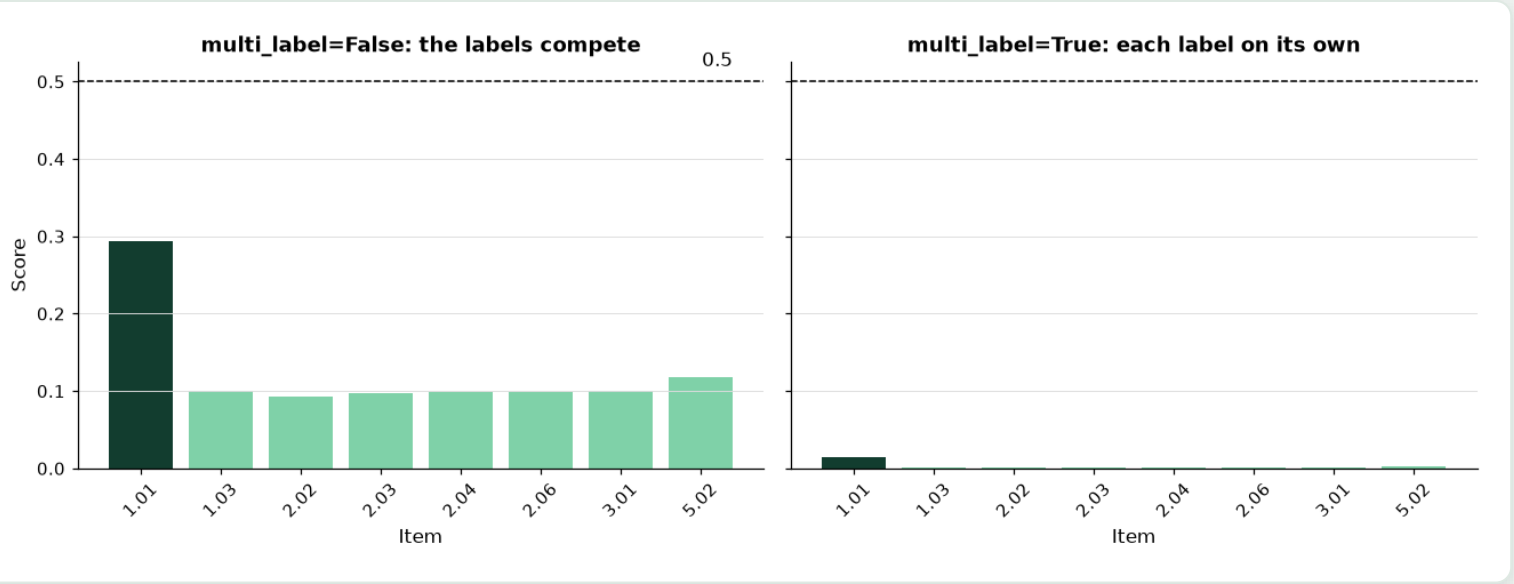
T = 1.0

**0.196**

T = 2.0

**The order of the items never changes. The pipeline always uses 1; day 11 meets the setting in a language model's request.**

# multi_label true




**0.026**

the scores add up to, True

**0 / 0**

labels above 0.5, False / True

**True is for texts that can carry several labels. An 8-K section is filed under one item, so the course keeps False.**

# Every setting of the request

### candidate_labels (default (required))

The names the texts are compared with: a list, or one string of comma-separated labels. Each label becomes one hypothesis, so one pass of the model.

### hypothesis_template (default "This example is {}.")

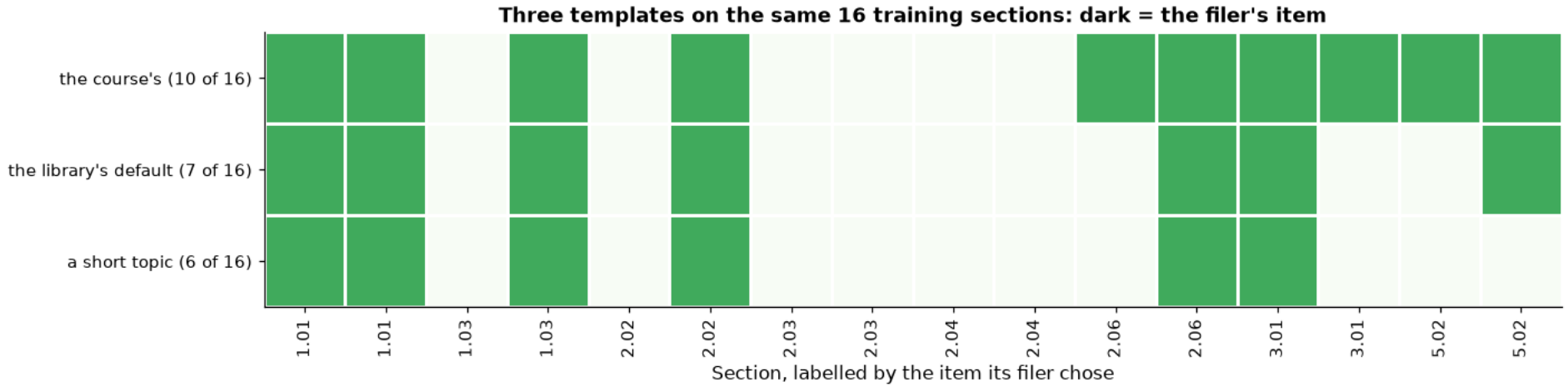
The sentence each label is put into; the braces mark where the label goes.

### multi_label (default False)

False: the labels compete, and a text's scores add up to 1. True: each label is scored alone, entailment against not_entailment.

**4 settings, the arguments the library's documentation lists; the three shaded change what the scores mean, and the course sets 3.**

# Three templates, sixteen sections




**10**

of 16, the course's

**7**

of 16, the library's default

**6**

of 16, a short topic

**Training rows only, a small sample: read the pattern, not the count.**

# Excel side by side

**Excel side by side (tested in Excel)**

| | |
|---|---|
| `=INDEX($B$1:$I$1,MATCH(MAX(B2:I2),B2:I2,0))` | the zero-shot item on 741 of 741 rows |
| `=SUMPRODUCT(--(K2:K742=J2:J742))/ROWS(K2:K742)` | 0.6302, within 1e-12 |
| `=COUNTIFS(K2:K742,"2.04",J2:J742,"2.04")` | 33 |

**No zero-shot or sentence model in Excel: the scores come from PyTorch.**

# Save the day: git diff results.csv

```
git diff results.csv
+zero_shot,validation,0.5951,0.6262,0.6400,0.3300,741
+embedding_logistic,validation,0.8882,0.8934,0.8700,0.8500,741
```

**pytest: 40 passed. A log with 4 decimals and LF endings diffs cleanly.**

# The AI check

The docstring is the prompt. The code that comes back runs, and its number looks fine.

1

## Docstring

Units, conventions, and what the function must never do: what you asked for.

2

## Code back

Written for the course in the style of an AI answer. Read it.

3

## Test first

Assert what the docstring promises. Run it: it fails.

4

## Fix

Find the rule it broke, fix it, run the test again.

**Today: sorted scores under fixed labels: all 741 sections assigned 1.01, macro F1 0.0297.**

*Code that runs is not the same as code that is right.*

# What you can do now

- One inference pass per label
- Read every score by its label's name
- Wording and template on training rows; temperature by hand
- Check a cached output with a live sample
- Agreement is not accuracy

**Week 2, Day 3** Clustering 10-K risk factors